EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A many-to-many relationship can always be represented correctly with just a single FK on one of the two entity tables.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

DO NOW — RECALL

Lesson 135 · Entity Relationships: One-to-Many and Mar

Name: _____ Class: _____ Date: _____

1. What is the role of a foreign key?

2. In Big O notation, what is binary search worst-case complexity?

3. What does ROM stand for?

4. Which is removed first from a FIFO queue?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It references a candidate/primary key in a related table.

2. Model answer

O(log n) for sorted random-access data.

3. Model answer

Read-only memory.

4. Model answer

The longest-waiting item at the front.

By the end, you can…

- Find objects with independent identity, not every noun.
- One student takes many courses, and each course has many students.
- Add Enrolment table referencing StudentID and CourseID.

Vocabulary — know these terms

entity — An identifiable class of thing stored in database.

relationship — Association between entities.

cardinality — How many related instances are possible.

junction table — Table breaking a many-to-many relation into two one-to-many relations.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate why a field containing comma-separated ClubIDs is poorer than a Membership junction table. [8]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain the difference between one-to-many and many-to-many relations with examples. [6]

Question 2

Design tables for Author and Book when each book has multiple authors. [6]

NEW LEARNING

Read, annotate and explain

Identify entities

Find objects with independent identity, not every noun.

Set cardinality

One student takes many courses, and each course has many students.

Resolve M:N

Add Enrolment table referencing StudentID and CourseID.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Represent students enrolling on several courses, with many students per course.

Student↔Course is M:N

Find the given data and name the required result.

Create Enrolment bridge

Explain why this intermediate stage is necessary.

Two one-to-many links

Student(StudentID PK) and Course(CourseID PK) form many-to-many. Resolve with Enrolment(StudentID FK, CourseID FK, EnrolledOn), with composite PK (StudentID, CourseID) if enrolment is unique per pair.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve